

BEFA (Unit -5)

Topic – 1

Introduction to Ratio Analysis & Types of Ratios

Ratio analysis is referred to as the study or analysis of the line items present in the financial statements of the company. It can be used to check various factors of a business such as **profitability, liquidity, solvency** and efficiency of the company or the business.

Ratio analysis is mainly performed by external analysts as financial statements are the primary source of information for external analysts.

The analysts very much rely on the current and past financial statements in order to obtain important data for analysing financial performance of the company. The data or information thus obtained during the analysis is helpful in determining whether the financial position of a company is improving or deteriorating.

Liquidity:

- Liquidity refers to how well the firm is in a position to meet its short term commitments such as payment of salaries, taxes, and so on.

Profitability:

- Profitability refers to how capably the firm is conducting its business operations in a profitable manner.

Solvency:

- Solvency ratio refers to the firm's position to meet its long-term commitments such as repayment of long-term loans, and so on.
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What Are the Objectives of Ratio Analysis?

- Provides analysis of profitability, liquidity, the solvency of the business.
- Traces the area which seeks more attention
- Helps in making estimates

Supports in comparing the performance with best industry standards

Use of Ratio Analysis

Ratio analysis is useful in the following ways:

1. Comparing Financial Performance: One of the most important things about ratio analysis is that it helps in comparing the financial performance of two companies.
2. Trend Line: Companies tend to use the activity ratio in order to find any kind of trend in the performance. Companies use data from financial statements that is collected from financial

statements over many accounting periods. The trend that is obtained can be used for predicting the future financial performance.

3. Operational Efficiency: Financial ratio analysis can also be used to determine the efficiency of managing the asset and liabilities. It helps in understanding and determining whether the resources of the business is over utilized or under-utilized.

Categories of Ratio Analysis

I) Liquidity Ratios

II) Activity Ratios

III) Capital Structure Ratios (Leverage Ratios)

IV) Profitability Ratios

I) Liquidity Ratios

A liquidity ratio is a type of financial ratio used to determine a company's ability to pay its short-term debt obligations. The metric helps determine if a company can use its current, or liquid, assets to cover its current liabilities.

Types of Liquidity Ratio

There are following types of liquidity ratios:

1. Current Ratio or Working Capital Ratio
2. Quick Ratio also known as Acid Test Ratio

Current Ratio or Working Capital Ratio

The current ratio is a measure of a company's ability to pay off the obligations within the next twelve months. This ratio is used by creditors to evaluate whether a company can be offered short term debts. It also provides information about the company's operating cycle. It is also popularly known as Working capital ratio. It is obtained by dividing the current assets with current liabilities.

$$\text{Current ratio} = \text{Current Assets} / \text{Current Liabilities}$$

A higher current ratio around two(2) is suggested to be ideal for most of the industries while a lower value (less than 1) is indicative of a firm having difficulty in meeting its current liabilities.

Quick Ratio or Acid Test Ratio

Quick ratio is also known as Acid test ratio is used to determine whether a company or a business has enough liquid assets which are able to be instantly converted into cash to meet short term dues. It is calculated by dividing the liquid current assets by the current liabilities

$$\text{Quick Ratio} = \text{Current Assets} - (\text{Stock} + \text{prepaid expenses}) / \text{Current liabilities}$$

The ideal quick ratio should be one(1: 1) for a financially stable company.

Current Assets:

- Stock
- Debtors
- Bills Receivable
- Cash at bank
- Cash in hand
- Prepaid expenses
- Income yet to received, so on..

Current Liabilities:

- Creditors
- Bank over draft (less than 1 year)
- Bills payable
- Out standing expenses
- Incomes received in advance
- All provisions
- Dividends payable, so on....

1. Calculate the different liquidity ratios from the following particulars

Particulars	Rs
Inventory	1,50,000
Cash	50,000
Sundry Debtors	3,00,000
Creditors	3,50,000
Bills receivables	30,000
Bank overdraft	30,000

$$\text{Current Ratio} = \text{Current Assets} / \text{Current Liabilities}$$

Current Assets = Sundry Debtors + Inventories + Cash-in-hand + Bills Receivable

Current Assets= 300,000 + 150,000+ 50,000+ 30,000= 530,000

Current Liabilities = Creditors + Bank Overdraft

Current Liabilities = 350,000 + 30,000 = 380,000

Current Ratio = 530,000 / 400,000 = 1.3 : 1

*For every one rupee of current liabilities there is Rs 1.3 worth current assets

Hence the liquidity position of the firm is not satisfactory as it is less than the standard i.e. 2 : 1

Quick Ratio or Acid Test Ratio = Quick Assets / Current Liabilities

Quick Assets = Current Assets – Inventories

Quick Assets = 530,000 - 150,000 = 380,000

Quick Ratio or Acid Test Ratio = 380,000 / 380,000 = 1:1

*Since this is equal to the standard of 1: 1, short term liquidity position of the company is satisfactory

2) Calculate Current Ratio, Quick Ratio and from the figures given below:

Particulars	Rs
Inventory	30,000
Prepaid Expenses	2,000
Other Current Assets	50,000
Current Liabilities	40,000
12% Debentures	30,000
Accumulated Profits	10,000
Equity Share Capital	1,00,000
Non-current Investments	15,000

Solution

(i) Current Assets = Inventory + Prepaid Expenses + Other Current Assets

= 30,000 + 2,000 + 50,000 = 82,000

Current Liabilities = 40,000

$$\begin{aligned}\text{Current Ratio} &= \frac{\text{Current Assets}}{\text{Current Liabilities}} \\ &= \frac{82,000}{40,000} = 2.05 : 1\end{aligned}$$

*For every one rupee of current liabilities there is Rs 2.05 worth current assets

Hence the liquidity position of the firm is satisfactory as it is slight more than the standard i.e. 2 : 1

(ii) Liquid Assets = Current Assets – Inventory – Prepaid Expenses

$$= 82,000 - 30,000 - 2,000 = 50,000$$

$$\text{Quick Ratio} = \frac{\text{Liquid Assets}}{\text{Current Liabilities}} = \frac{50,000}{40,000} = 1.25 : 1$$

*Since this is also above the standard of 1: 1, short term liquidity position of the company is satisfactory

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Topic – 2

Types of Ratios

II) Activity Ratios

- Activity ratios express how active the firm is in terms of selling stocks, collecting its receivable and paying its creditors.
- These are 3 types: a) Inventory turnover ratio b) Debtors Turnover ratio c) Creditors turn over ratio
- An activity ratio is a type of financial metric that indicates how efficiently a company is leveraging the assets on its balance sheet, to generate revenues and cash.
- Commonly referred to as efficiency ratios, activity ratios help analysts gauge how a company handles inventory management, which is key to its operational fluidity and overall fiscal health.

Inventory Turnover Ratio

- It is also called Stock turnover ratio.
- It indicates the number of times the average stock is being sold during a given accounting period.
- It establishes the relation between the cost of goods sold during a given period and the average amount of inventory outstanding during that period.
- The higher the inventory turnover ratio, the better is the performance of the firm selling its stocks
- It helps in determining the liquidity of the firm by giving the rate at which inventories are converted into sales and then into cash.

$$\text{Inventory Turnover Ratio} = \text{Cost of the goods}/\text{Average inventory}$$

- Where cost of goods sold = sale – Gross profit
- Average inventory is the average of opening stock at the beginning of the year and the closing stock at the end of the year, that is

$$\text{Average inventory} = \text{Opening stock} + \text{closing stock}/2$$

1) Calculate inventory turnover Ratio./

A firm sold good worth Rs 5, 00,000 and its gross profit is 20% of sales value. The inventory at the beginning of the year was Rs. 16,000 and at the end of the year were 14,000. Compute inventory turnover ratio and also the inventory holding period.

Sol) Calculation of inventory turnover ratio, we need cost of goods sold (COGS) and Average stock.

Cost of goods sold = sales-gross profit

Gross profit = 20% of sales value (20% of 5, 00,000 = 1, 00,000)

COGS = 5, 00,000 – 1, 00,000 = **4, 00,000**

$$\begin{aligned}\text{Average Inventory} &= (16,000 + 14,000) / 2 \\ &= 15,000\end{aligned}$$

$$\begin{aligned}\text{Inventory turnover ratio} &= \text{cost of goods sold} / \text{average inventory} \\ &= 4,00,000 / 15,000 \\ &= \mathbf{26.66 \text{ times}}\end{aligned}$$

This means that during the year, the average stock is being sold 26.66 times

$$\begin{aligned}\text{b) Inventory holding period} &= 365 \text{ days} / \text{inventory turnover ratio} \\ &= 365 / 26.66 \\ &= \mathbf{13.69 \text{ days or 14 days approximately}}\end{aligned}$$

Debtors Turn Over Ratio

- It reveals the no. of times the average debtor are collected during a given accounting period.
- In other words, it shows how quickly the firm is in a position to collect its debts.
- In case, the firm is not in a position to collect its debts, to meet the working capital requirement, it has to borrow paying interest.
- A higher debtors turnover ratio explains that the firm is efficient in collecting its debts whereas lower ratio signifies its inefficiency

$$\mathbf{\text{Debtors Turnover Ratio} = \text{Credit sales} / \text{Average debtors}}$$

Debt Collection Period:

Debt collection period refers to the time taken to collect the debts.

$$\mathbf{\text{Debt Collection period} = 365 \text{ days} / \text{debtors turnover ratio}}$$

The lesser time, more is the efficiency of the firm vice versa.

Problem;

A firm's sale during the year was Rs. 4,00,000 of which 60% were on credit basis. The balance of debtors at the beginning and end of the year were 25,000 and 15,000 respectively. Calculate debtor's turnover ratio of the firm. Also find out debt collection period.

$$\begin{aligned}\text{Sol) Credit sales} &= 60\% \text{ of } 4,00,000 \\ &= \mathbf{2,40,000}\end{aligned}$$

$$\begin{aligned}\text{Average debtors} &= (\text{Opening balance of debtors} + \text{Closing balance of debtors}) / 2 \\ &= (25,000 + 15,000) / 2 \\ &= \mathbf{20,000}\end{aligned}$$

Calculation of Debtors turnover ratio = 2, 40,000/20,000

= 12 times

In this example, the firm is collecting its average debtors 12 times during the given accounting period.

Debt collection period = 365days/ debtors turnover ratio

= 365/12

= 30.14 days

On an average, the firm is taking around 31 days to collect its debts

Creditors Turnover Ratio

- It reveals the number of times the average creditors are paid during a given accounting period.
- It is necessary to keep close monitoring of payment schedules because it directly affects the working capital position.
- In case the firm is not in a position to pay its creditors, it will affect the good will or further supplies may be cut off.

Creditors Turn Over Ratio = Credit purchases/ Average Creditors

Creditors payment period = 365 days/ Credit turnover ratio

Calculate C.T.O.R & Creditors Payment Period

Total net credit purchases for year 2021: Rs12,50,000

Accounts payable balance January 1, 2021: Rs2,08,000

Accounts payable balance December 31, 2021: Rs2,24,000

Average accounts payable = (Rs 2,08,000 + Rs 2,24,000) / 2 = Rs2,16,000

AP turnover ratio (or) creditors T.O.R= Credit Purchases / Average Creditors

=Rs12,50,000 / Rs 2,16,000

= 5.8 times per year

- **Creditors Payment Period is calculated by using C.T.O.R**
- One-year formula: 365 days / AP turnover ratio = Days payable outstanding
- One-quarter formula: 90 days / AP turnover ratio = Days payable outstanding
- One-month formula: 30 days / AP turnover ratio = Days payable outstanding
- Converting the AP turnover ratio from the one-year example used above:
- 365 / 5.8 = 63 Days payable outstanding
- Ie.. Once in every 63 days credit payments are done to its creditors.

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Topic – 3

Types of Ratios

Capital Structure Ratios/ Leverage Ratios

- Capital structure or leverage ratio is defined as ‘the financial ratio which focuses on the **long-term solvency of the firm**’.
- Long term solvency reflects its ability to meet its long-term commitments such as payments of interest periodically without fail, repayment of principal as and when due.
- Types of capital structure ratios are: Debt-equity ratio, Interest coverage ratio, Ratios ‘of proprietors funds to Total assets

Debt – Equity Ratio

- It is the ratio between outsiders’ fund (debt) and insider’s funds (equity).
- This is used to measure the firm’s obligations to creditors in relation to the owner’s funds.
- It is a measure of solvency
- The yard stick for this ratio is 1:1

$$\text{Debt Equity ratio} = \text{Debt/Equity}$$

- Debt or outsider’s fund include debentures, bonds, long-term loans, and so on.
- Share holders’ fund or equity here includes share capital (both preference and equity), reserves (both general and specific), retained earnings, and such others.
- Equity does not only mean equity share capital.
- Equity here is interpreted as ‘insiders’ funds’. ‘Debt here means only long-term debt.

Problem

Calculate Debt Equity Ratio from the following information:

Particulars	Rs
Equity Share Capital	3,00,000
Preference Share Capital	1,00,000
General Reserve	1,30,000
Securities Premium	40,000
10 % Debentures	4,50,000
Profit & Loss Balance	70,000
Loan from Bank	3,50,000
Current Liabilities	20,000

Solution

Debt Equity Ratio = Debt/Equity or Long term Debts Shareholder's Funds

Long-term Debts = 10% Debentures + Loan from Bank

$$= \text{Rs. } 4,50,000 + \text{Rs. } 3,50,000 = \text{Rs. } 8,00,000$$

Shareholder's Funds = Equity Share Capital + Preference Share Capital + General Reserve +

Securities Premium + Profit & Loss Balance

$$= \text{Rs. } 3,00,000 + \text{Rs. } 1,00,000 + \text{Rs. } 1,30,000 + \text{Rs. } 40,000 + \text{Rs. } 70,000$$

$$= \text{Rs. } 6,40,000$$

$$\text{Debt Equity Ratio} = 8,00,000/6,40,000 = 1.25:1$$

Interest Coverage Ratio

- Interest coverage ratio is calculated to judge the firm's capacity to pay the interest on debt it borrows.
- It gives an idea of the extent the firm's earning may contract before it is unable to pay interest payments out of current earnings.
- It is very important ratio for the financial institutions to judge the ability of the borrow to service the loan from the current year's profits
- The higher the ratio, better it is.

Interest coverage ratio = Interest and Taxes/Fixed interest charges

The more the number of times of coverage, better is the solvency position of the borrower.

Problem

The earnings before interest and taxes (EBIT) of a company is Rs. 5,60,000. Its fixed commitments include payment of 10% on 7000 debentures of Rs. 100 each. It is subject to tax of 30% p.a.

Sol) Net profit before interest and taxes = Rs.5,60,000

Fixed interest charges on the debentures = (7000*100)

$$= \text{Rs. } 7,00,000 * 10\%$$

$$= \text{Rs } 70,000.$$

Interest Coverage Ratio = 5,60,000/70,000

$$= 8 \text{ times}$$

- Interest coverage ratio of 8 times means that the net profit earning are 8 times to the fixed interest charge payable during the year.
- The more the number of times the coverage the safer is the investment.

Ratio of Proprietors' fund to Total Assets

- This establishes the relationship between proprietors' funds and the total assets.
- Here the total assets include the tangible fixed assets plus current asset.
- As a guideline a ratio of around 0.5:1 or 50% is considered as the minimum desirable.

$$\text{Ratio of proprietors' funds to Total Assets} = (\text{Proprietors Funds} / \text{Total Assets}) \times 100$$

Problem

Company XYZ has the following details:

- Proprietors' Fund (Shareholders' Equity) = 10,00,000
- Total Assets = 25,00,000

Calculate the Proprietors' Fund to Total Assets Ratio.

Sol) Proprietors' Fund to Total Assets Ratio = Proprietors fund/Total Assets $\times 100$

$$\begin{aligned}\text{Proprietors' Fund to Total Assets Ratio} &= 10,00,000 / 25,00,000 \times 100 \\ &= 0.4 \times 100 = 40\%\end{aligned}$$

This means that 40% of the company's assets are financed by owners' equity, and the remaining 60% is financed by liabilities (debts).

Ratio of Fixed Assets to Proprietors' Fund.

- This ratio Explains whether the fixed assets have been bought from the proprietors funds or not.
- It is not safe to use short-term finance to buy long-term assets because when the borrowings are to be repaid, there may be a problem, as fixed assets cannot be readily converted into cash.

$$\text{Ratio of fixed assets to proprietors' funds} = (\text{fixed assets} / \text{proprietors' funds}) \times 100$$

Problem

Company ABC has the following details:

- **Fixed Assets** = 12,00,000
- **Proprietors' Fund** (Owner's Equity) = 15,00,000

Calculate the Ratio of Fixed Assets to Proprietors' Fund.

Sol) Ratio of Fixed Assets to Proprietors' Fund = Fixed Assets/proprietors' fund $\times 100$

$$\begin{aligned}&= 12,00,000 / 15,00,000 \times 100 \\ &= 80\%\end{aligned}$$

Higher Ratio: A higher ratio indicates that a larger portion of the owners' equity is being used to acquire fixed assets.

Ratio of Current Assets to Proprietors' Funds

- A higher ratio of current assets to proprietors' funds is considered as financial strength to the business.
- It is necessary to hold adequate funds in working capital to generate profits.

$$\text{Ratio of Current Assets to Proprietors' fund} = (\text{Current Assets} / \text{Proprietors' fund}) \times 100$$

Problem:

Company XYZ has the following details:

Current Assets = 8, 00,000

Proprietors' Fund (Owner's Equity) = 12, 00,000

Calculate the Ratio of Current Assets to Proprietors' Fund.

Sol) Ratio of Current Assets to Proprietors' fund = (Current Assets/Proprietors' fund)*100

$$= (8, 00,000/12,00,000)*100$$

$$=66.67\%$$

A higher ratio indicates a larger portion of the owner's equity is invested in current assets, suggesting the company is more liquid and can meet its short-term obligations more easily.

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Topic – 4

Types of Ratios

Profitability Ratios

- It throws light on how well the firm is organizing its activities in a profitable manner.
- The owner expect reasonable rate of return on their investment.
- The firm should generate enough profits not only to meet the expectations of the owners, but also to finance the expansion activities.

Following 8 ratios commonly used to explain profitability:

- Gross profit ratio
- Net profit Ratio
- Operating ratio
- Return on investment (ROI)
- Earnings per share (EPS)
- Dividend yield
- Price/Earnings ratio (P/E ratio)
- Earning Power

Gross profit Ratio

- It is the ratio between gross profits to sales during a given period.
- It is expressed in terms of percentage.
- The higher the gross profit ratio, the better it is.

$$\text{Gross profit Ratio} = (\text{Gross profit/Sales}) \times 100$$

Suppose the net sales is 50,000 for a firm and cost of goods sold is Rs.20,000. the gross profit is calculated as below:

$$\text{Sol) Gross profit} = (30,000/50,000) \times 100$$

$$= 60\%$$

$$\text{Gross profit} = \text{Net sales} - \text{cost of goods sold}$$

$$= 50,000 - 20,000$$

$$= 30,000$$

In other words, 60% of its sales is gross profit

Net Profit Ratio:

- It is the ratio between net profit after taxes and net sales.
- It indicates what portion of sales is left to the owners after operating expenses.
- Non-operating income such as interest on investments, gain on sale of fixed assets and so on are added to the operating profit and non-operating expenses such as loss on sale of fixed assets and so on deducted from such profit.
- This is the net profit after adjusting non-operating income and non-operating expenses.

$$\text{Net profit ratio} = (\text{Net profit after taxes} / \text{Net sales}) * 100$$

Suppose the net sales are Rs 50,000 for a firm and cost of goods sold is Rs. 20,000. The details of expenses are as given below:

Administration expenses	Rs. 4,000
Selling and Distribution Expenses	Rs. 4,000
Loss on sale of fixed assets	Rs. 3,000
Interest on Investment	Rs. 2,000
Taxes @20%	

Sol:

Sales	50,000
- Less cost of goods sold	-20,000

Gross profit		30,000
- Less selling and distribution expenses	-4,000	}
- Administration expenses	-4,000	
		-8,000

Net profit	22,000
+ Interest on investments (non-operating income)	+2,000

24,000

- Loss on sale of assets	- 3,000
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21,000

- Taxes @20%	- 4,200
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Net profit after Taxes	16,800
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Operating Ratio

- Operating ratio is the ratio between costs of goods sold plus operating expenses and the net sales.
- The higher the operating ratio, the lower is the profitability and vice versa.
- Where operating expenses = (cost of goods sold + administration expenses+ Selling and distribution expenses)

$$\text{Operating ratio} = (\text{operating expenses/net sales}) * 100$$

Problem:

Company ABC has the following details:

Net Sales = 20, 00,000

Operating Expenses (including COGS, selling, and administrative expenses) = 14, 00,000

Calculate the Operating Ratio.

Solution:

$$\begin{aligned}\text{Operating Ratio} &= (\text{Operating Expenses/Net Sales}) * 100 \\ &= (14, 00,000 / 20, 00, 000) * 100 \\ &= 70\%\end{aligned}$$

Return on Investment (ROI)

- ROI is one of the very important parameters affecting business plans.
- The profitability of the firm is measured in terms of return on investment.
- The term investment may refer to total assets, capital employed or owners' equity.

$$\text{ROI} = \text{Net Profit after Taxes/Total Investments}$$

Some of the important ratios under this head are:

- a) Return on Capital Employed (ROCE)
- b) Return on Equity (ROE)

$$\text{ROCE} = \text{Adjusted Net Profits/Capital Employed}$$

- This is only satisfactory measure which reveals the overall performance of a firm in terms of profitability.

$$\text{ROE} = (\text{Net profit-Dividend payable to preference shareholder}) / \text{Equity share capital}$$

Earnings per share (EPS)

- EPS is the relationship between net profits and the number of shares outstanding at the end of the given period.
- This is can be compared with previous years to provide a basis for assessing the company's performance.

$$\text{EPS} = (\text{Net profit after taxes/number of shares outstanding})$$

Given that the number of shares is 10,000 and net profit after taxes for a given period is Rs. 4, 50,000; the EPS can be calculated as follows:

$$\begin{aligned}\text{EPS} &= 4, 50,000/10,000 \\ &= \text{Rs. } 45\end{aligned}$$

The higher the EPS is, the more is likely to be the demand for the shares of the company. However, it is to be noted that EPS is one of the many affecting the demand for a given share.

Dividend Yield

- Yield refers to the amount of total return the investor will receive for a given period of time for the amount of his investment.
- Dividend yield refers to the percentage return on the price paid for shares

Dividend yield = (Nominal or face value of share/cost or market price of the share)*% dividend p.a.

Given that current market price of a share Rs. 300; face value of the share is Rs.100; * of dividend declared is 20%, then the yield is

$$\begin{aligned}\text{Dividend Yield} &= (100/300)*20 \\ &= 1/3*20 \\ &= 6.66 \text{ p.a.}\end{aligned}$$

The higher yield reflects that the investments are riskier and lower the yield , safer are the investments.

- In general, yield and risk are inversely proportional.

Price Per Earing Ratio

Given that market price of a share is Rs. 340 and EPS is 10, calculate P/E ratio.

$$\text{Price/Earnings Ratio} = (\text{Market price per share/Earning per share})$$

$$\begin{aligned}\text{Sol) Price/Earnings Ratio} &= (\text{Market price per share/Earning per share}) \\ \text{EPS} &= (340/10) \\ &= 34\end{aligned}$$

Thus a share with market price of Rs. 340 and EPS of Rs. 10 would have a P/E ratio of Rs.34

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Topic – 5

Funds Flow Analysis

Introduction

- The term funds, we generally mean cash.
- But from the point of view of accountants and finance managers, ‘funds’ means **working capital or net working capital**.
- In the context of funds flow statement, when we use term funds, it means working capital only.

Objectives of Fund flow statement:

- The funds flow statement is prepared to identify different sources and application of funds.
- In other words, it explains the sources of raising funds and how the same has been utilized .
- The funds flow statement answers the following specific questions:

1) How could the firm pay dividends at a higher rate despite declining profits

in the year?

2) Is the firm solvent or not?

3) How the company has financed its fixed assets?

4) Why does the company find difficult to pay dividends though it has made good profits?

Current Assets and Current Liabilities Listed

Current Assets:

- Cash in hand
- Cash at bank
- Bills receivable
- Sundry debtors
- Marketable/temporary investments
- Advances (short term)
- Closing inventory or finished goods/ work in progress/raw materials
- Prepaid expenses
- Accrued income

Current Liabilities:

- Bank overdraft

- Sundry creditors
- Bills payable
- Outstanding expenses
- Dividends payable
- Taxes payable
- Provision for taxation
- Provision for dividends
- Unearned income

Funds Flow Statement

Preparation of funds flow statement involves 3 stages:

1. Statement or schedule of changes in working capital
2. Funds from operations
3. Funds flow statement of sources and application of funds

Following are the rules for the preparation of Statement of changes in working capital:

- Increase in current asset will result in increase in working capital.
- Decrease in current asset will result in decrease in working capital.
- Decrease in current liability will result in increase in working capital.
- Increase in current liability will result in decrease in working capital.

We find a ***direct relationship between current assets and working capital*** and an ***inverse relationship between current liabilities and working capital***

$$\text{Working Capital} = \text{Current Assets} - \text{Current Liabilities}$$

Proforma of Statement of Changes in Working Capital

Statement of Changes in Working Capital (for the year ended _____)

Particulars	Previous Year	Current Year	Working Capital Changes	
			Increase	Decrease
Current Assets:	(₹)	(₹)	(₹)	(₹)
Cash in Hand				
Cash at Bank				
Short-term Investment (Marketable Securities)				
Sundry Debtors				
Bills Receivable				
Accrued Income				
Prepaid Expenses				
Stock-in-trade				
Other Current Assets				
(A)	xxxx	xxxx		
Current Liabilities:				
Sundry Creditors				
Bills Payable				
Outstanding Expenses				
Income received in advance				
Bank Overdraft				
Other Current Liabilities				
(B)	xxxx	xxxx		
Working Capital (A - B)				
Increase/Decrease in Working Capital			xxxxxx	xxxxxx

Problem:

Example 1

From the following Balance Sheet of XYZ Limited, prepare a Statement of changes in working capital.

<i>Capital and liabilities</i>	<i>31.12.2016</i> (₹)	<i>31.12.2017</i> (₹)
Share capital	6,00,000	7,50,000
Sundry creditors	1,60,000	1,00,000
Bills Payable	52,000	40,000
Profit and loss account	28,000	62,000
	8,40,000	9,52,000
<i>Assets</i>		
Machinery	1,40,000	2,00,000
Stock	2,42,000	2,72,000
Debtors	3,62,000	3,40,000
Cash	96,000	1,40,000
	8,40,000	9,52,000

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Solution**Statement of Changes in Working Capital**

	<i>2016</i>	<i>2017</i>	<i>Increase in</i> <i>working capital</i>	<i>Decrease in</i> <i>working capital</i>
	(₹)	(₹)	(₹)	(₹)
<i>Current assets</i>				
Cash	96,000	1,40,000	44,000	
Debtors	3,62,000	3,40,000		22,000
Stock	2,42,000	2,72,000	30,000	
	<u>7,00,000</u>	<u>7,52,000</u>		
<i>Current liabilities</i>				
Sundry creditors	160,000	1,00,000	60,000	
Bills payable	52,000	40,000	12,000	
	<u>2,12,000</u>	<u>1,40,000</u>		
Working capital	4,88,000	6,12,000		
Net increase in working capital	1,24,000			1,24,000
	<u>6,12,000</u>	<u>6,12,000</u>	1,46,000	1,46,000

Example 2

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Funds from Operations

- Funds from operations will be calculated by making adjustments of several items to the net profit.

Items to be added to net profit:

- Depreciation . Amortization of assets . Loss on sale of fixed assets
- Discount on issue of shares/debentures . Interim Dividend paid., premium on redemption of preference shares and debentures.
- Transfer of profit to general reserve or reserve fund.

Items to be deducted from net profit

- Gain on sale of fixed assets
- Income from investments
- The funds flow statement contains sources of funds and uses of applications of funds
- If the sources of funds are more than the uses or application of funds, then there is increase in working capital and vice versa.
- Funds from operations will be calculated by making adjustments of several items to the net profit.

Items to be added to net profit:

- Depreciation . Amortization of assets . Loss on sale of fixed assets
- Discount on issue of shares/debentures . Interim Dividend paid., premium on redemption of preference shares and debentures.
- Transfer of profit to general reserve or reserve fund.

Items to be deducted from net profit

- Gain on sale of fixed assets
- Income from investments
-

Problem

Ramco Cements presents the following information and you are required to calculate funds from operations

PROFIT AND LOSS ACCOUNT			
	Rs.		Rs.
To Operation Expenses	1,00,000	By Gross Profit	2,00,000
To Depreciation	40,000	By Gain on Sale of Plant	20,000
To Loss on Sale of Building	10,000		
To Advertisement Suspense Account	5,000		
To Discount Allowed	500		
To Discount on Issue of Shares written off	500		
To Goodwill written off	12,000		
To Net Profit	52,000		
	<u>2,20,000</u>		<u>2,20,000</u>

Solution

Calculation of funds from operation

Net profit given	Rs	Rs
ADD: Non-fund or Non- operating items which have been debited to P& L A/c		52,000
• Depreciation		
• Loss on sale of building	4,0000	
• Advertising suspense account	10,000	
• Discount on issue of shares written off	5,000	
• Goodwill written off	500	
	<u>12,000</u>	<u>67,500</u>
LESS: : Non-fund or Non- operating items which have been credited to P& L A/c		1,19,500
Gain on sale of plant		
FUNDS FROM OPERATIONS-----	20,000	<u>(-) 20,000</u>
-----		<u>99,500</u>
-		

Alternative method

Adjusted P & L A/c

	Rs		Rs
To depreciation	40,000	By opening balance	---
To Loss on sale of building	10,000	Gain on sale of plant	20,000
To Advertising suspense account	5,000		
To Discount on issue of shares written off	500		
To Goodwill written off	12,000	By Funds from operations----->	<u>99,500</u>
To closing balance	<u>52,000</u>		
(Net Profit given)	<u>1,19,000</u>		<u>1,19,500</u>

Funds Flow Statement

- The funds flow statement is a statement containing sources of funds and application or uses of funds.

Format of Fund Flow Statement

Sources / Inflow	Rs	Applications/Outflow	Rs
Funds from Operations	**	Funds lost in operations	**
Sale of fixed assets	**	Purchase of fixed assets	**
Sale of Investments	**	Purchase of investments	**
Issue of shares	**	Redemption of preference shares	**
Issue of debenture	**	Redemption of debenture	**
Loan taken	**	Loan repaid	**
Dividend Received	**	Dividend paid	**
Non-Operating income	**	Non – operating expenses	**
Decrease in working capital	**	Increase in working capital	**
	**		**

BEFA (Unit -5)

Topic – 6

Worked out Problems on Funds Flow Analysis

- A fund flow statement is a document that covers the inflows and outflows of funds.
- The funding sources and the use of funds in a given period will be included.
- Therefore, the reasons behind the change in a company's finances can be analysed.

It is one of the critical indicators that shows how funds are used. Finance analysts will be able to evaluate the fund flow of an organisation in due course under this statement. It is also known as the application of the Funds and the Statement of Sources, as this statement describes the movement of funds between different sources and their applications.

Format of Fund Flow Statement

Sources / Inflow	Rs	Applications/Outflow	Rs
Funds from Operations	**	Funds lost in operations	**
Sale of fixed assets	**	Purchase of fixed assets	**
Sale of Investments	**	Purchase of investments	**
Issue of shares	**	Redemption of preference shares	**
Issue of debenture	**	Redemption of debenture	**
Loan taken	**	Loan repaid	**
Dividend Received	**	Dividend paid	**
Non-Operating income	**	Non – operating expenses	**
Decrease in working capital	**	Increase in working capital	**
	**		**

Problem

Example 5

Given are the balance sheets of ABC Limited. You are required to prepare a Funds Flow Statement.

	As on 31 st March, 2017 (₹)	As on 31 st March, 2018 (₹)
Fixed assets at cost	7,00,000	8,70,000
Less: Depreciation	2,56,000	3,60,000
Net fixed assets (A)	4,50,000	5,10,000
Current asset		
Investments (temporary)	1,00,000	1,50,000
Closing inventory	18,15,000	19,00,000
Sundry debtors	13,15,000	13,87,000
Total of current assets (B)	32,30,000	34,37,000
Less: Current liabilities		
Bank overdraft	11,60,000	5,50,000
Trade creditors and provision	9,98,000	11,92,000
Proposed dividend	1,60,000	2,40,000
Total current liabilities (C)	23,18,000	19,82,000
Working capital (D) = (B) – (C)	9,12,000	14,55,000
Net fixed assets + Net Working Capital (A) + (D)	13,62,000	19,65,000
Represented by		
Ordinary share capital	7,56,000	10,00,000
General Reserve	2,60,000	3,80,000
Profit and Loss Account	3,52,000	4,85,000
8% Debentures	—	1,00,000
	13,62,000	19,65,000

Solution

Solution

Statement of changes in working capital

	2017 (₹)	2018 (₹)	Increase in working capital (₹)	Decrease in working capital (₹)
<i>Current assets</i>				
Investments (Temporary)	1,00,000	1,50,000	50,000	
Closing inventory	18,15,000	19,00,000	85,000	
Sundry debtors	13,15,000	13,87,000	72,000	
	<u>32,30,000</u>	<u>34,37,000</u>		

Bank overdraft	11,60,000	5,50,000	6,10,000	
Trade creditors and provision	9,98,000	11,92,000		1,94,000
Proposed dividend	1,60,000	2,40,000		80,000
	23,18,000	19,82,000		
Increase in working capital			8,17,000	5,43,000
				8,17,000

Funds from Operations

	Amount (₹)	Amount (₹)
Net profit in the current year (4,85,000 – 3,52,000)		1,33,000
Add: General reserve (3,80,000 – 2,60,000)		1,20,000
Depreciation during the year (3,60,000 – 2,50,000)		1,10,000
Funds from Operations		3,63,000

Funds Flow Statement

Sources	(₹)	Applications	(₹)
Funds from operations	3,63,000	Purchase of Fixed assets	1,70,000
Issue of shares	2,50,000	Increase in working capital	5,43,000
8% debentures issued	1,00,000		
	7,13,000		7,13,000

BEFA (Unit -5)

Topic – 7

Cash Flow Analysis

Introduction

- A statement of changes in cash flows that explains the various sources of cash and uses of cash during a given specific period is called **cash flow statement**.
- An analysis of cash inflows and cash outflows will enable the firm to make projections of cash flows (both inflows and outflows)
- Such projections will enable the firm to know about the availability of cash.
- A statement of changes in cash flow known as ‘cash flow statement’ explains the various sources of cash and the uses of cash.
- Categorizes cash flows into **operating**, **investing**, and **financing** activities.

Sources of cash

- Cash from operations
- Sales of fixed assets
- Issues of equity shares including share premium
- Issue of preference shares
- Issue of debentures
- Mortgage loans raised

Application or Uses of Cash

- Loss of cash from operations
- Purchases of fixed assets
- Redemption of preference shares
- Redemption of shares
- Repayment of long term loan
- The sources of cash reveals how the firm could get cash (i.e. cash and bank balances) and the application or uses reveals for what purpose cash outflow take place
- The opening cash and bank balance plus sources of cash less uses of cash should be equal to closing cash and bank balances

Computation of Cash from Operations

- Cash from operations can be calculated by adding **Net profit** for the year.

The following kinds of items will be added back to net profit

- ❖ Depreciation
- ❖ Amortization of fixed assets such as good and patents will written off.
- ❖ Loss on sale of fixed assets.
- ❖ Income tax provided for in profit and loss appropriation a/c.
- ❖ Proposed dividends provided for in P&L appropriation a/c.

After adding these items to the net profit, profit on sale of fixed assets, if any will deducted. The resulting amount is termed '**funds from operations**'

The following adjustments are to be made to the funds from operations

- Add increase in current liabilities (other than bank overdraft)
- Add decrease in current assets (other than cash and bank balances)
- Deduct decrease on current liabilities (other than bank overdraft)
- Deduct increase in current assets (other than cash and bank balance)

After making the above adjustments to the funds from operations.

Figure 19.1 is the proforma of statement of Cash from Operations.

Particulars	Amount (₹)
Net profit for the year	xxx
Add: Depreciation	xxx
General Reserve	xxx
Goodwill written off	xxx
Patents written off	xxx
Loss on sale of fixed assets	xxx
Income tax provided for in P & L A/c	xxx
Proposed dividends	xxx
Less: Profit on sale of fixed assets	xxx
Funds from Operations	xxx
Add: Increase in current liabilities (except bank overdraft)	xxx
Decrease in current assets (except cash and bank balances)	xxx
Less: Increase in current assets (except cash and bank balances)	xxx
Decrease in current liabilities (except bank overdraft)	xxx
Cash from Operations	xxx

Scanned with ACE Scanner

Problem

Example 1 After taking into consideration the under mentioned items, ABC Limited made a net profit of ₹2,00,000 for the year ended 31 March, 2018.

	(₹)
Depreciation on buildings	10,000
Depreciation on machinery	8,000
Preliminary expenses written off	10,000
Depreciation on furniture	5,000
Profit on sale of furniture	3,000
Provision for taxation	15,000
Goodwill written off	12,000

Compute cash from operations.

Scanned with ACE Scanner

Cash from operations

	Rs	Rs
Net Profit		2,00,000
ADD:		
Depreciation on Buildings	10,000	
Depreciation on machinery	8,000	
Depreciation on furniture	5000	
Preliminary expense	10,000	
Provision for tax	15,000	
Goodwillreturn off	<u>12000</u>	<u>60,000</u>
		2,60,000
LESS:		
Profit from sale of furniture		<u>(-) 3,000</u>
Funds / cash from operations		<u>2,57,000</u>

Difference between Funds Flow & Cash Flow Statement

Focus:

- **Funds Flow Statement:** Tracks the movement of **working capital** (current assets minus current liabilities).
- **Cash Flow Statement:** Focuses on the movement of **actual cash** inflows and outflows.

Content:

- **Funds Flow Statement:** Includes both cash and non-cash items, such as changes in assets, liabilities, and equity.
- **Cash Flow Statement:** Only includes **cash transactions**, excluding non-cash activities

Objective:

- **Funds Flow Statement:** Analyzes how funds (capital) are raised and used, focusing on the company's financial structure.
- **Cash Flow Statement:** Assesses the company's **liquidity** and ability to meet short-term obligations.

Scope:

- **Funds Flow Statement:** Deals with long-term financial changes and overall financial health.
- **Cash Flow Statement:** Focuses on short-term cash position and immediate cash needs.

Components:

- **Funds Flow Statement:** Focuses on changes in working capital, long-term liabilities, and equity.
- **Cash Flow Statement:** Categorizes cash flows into **operating, investing, and financing** activities.

Illustration:1

The summarized balance sheet of Bhadresh Ltd. as on 31.12.05 and 31.12.2006 are as follows:

Liabilities	2005	2006	Assets	2005	2006
Share capital	4,50,000	4,50,000	Fixed asset	4,00,000	3,20,000
General Reserve	3,00,000	3,10,000	Investment	50,000	60,000
P&I A/c	56,000	68,000	Stock	2,40,000	2,10,000
Creditors	1,68,000	1,34,000	Debtor	2,10,000	4,55,000
Tax provision	75,000	10,000	Bank	1,49,000	1,97,000
Mortgage loan	-	2,70,000			
	10,49,000	12,42,000		10,49,000	12,42,000

Additional Details:

1. Investment costing Rs.8,000 were sold for Rs.8,500
2. Tax provision made during the year was Rs.9,000
3. During the year part of fixed assets costing Rs10,000 was sold for Rs12,000 and the profit was included in P & L A/c. You are required to prepare cash flow statement for 2006.

Solution:**Cash flow statement for the year ended 31.12.2006**

Particular	Rs.	Rs.
1.Cash flows from operating activities:		
Net profit before tax(Rs.28,500 in case Profit on sale On Investment & Fixed Asset not considered)	31,000	
Adjustment for:		
Dep.	70,000	
Profit on sale of investment	-(500)	
Profit on sale of Fixed assets	-(2,000)	
Decrease in stock	30,000	
Decrease.in creditor	-(34,000)	
Increase. in debtor	-(2,45,000)	
Income tax paid	-(74,000)	
Net cash from operating activities (A)		(2,24,500)
2. Cash flows from investing activities:		
Investment purchased	(18,000)	
Sale of investment	8,500	
Sale of Fixed assets	12,000	
Net cash from investing activities (B)		2500
3.Cash flows from financing activities:		
Mortgage loan taken		2,70,000
Net cash from Investments (C)		
Net Cash Flow from all activities (A + B + C)		48,000
Add:		
Opening cash balance		<u>1,49,000</u>
Closing cash balance		<u>1,97,000</u>

Adjusted P & L A/c

Particulars	Rs	Particulars	Rs
To provision for tax	9,000	By Balance b/d	56,000
To provision for G	10,000	By profit on sale of investment	500
To Balance c/d	68,000	By profit on of fixed assets sale	2,000
		By adjusted profit (bal/fig)	<u>28,500</u>
	<u>87,000</u>		<u>87,000</u>

Fixed Assets A/c

Particulars	_Rs	Particulars	_Rs
To bal b/d	4,00,000	By bank a/c	12,000
To profit and loss a/c	2,000	By depreciation(bal/fig)	70,000
		By bal c/d	<u>3,20,000</u>
	4,02,000		4,02,000

Provision for tax A/c

Particulars	_Rs	Particulars	_Rs
To bank (tax paid) (bal/fig)	74,000	By bal b/d	75,000
T0 Bal c/d	10,000	By p & l a/c (provision)	9,000
	84,000		<u>84,000</u>

Investment A/c

Particulars	_Rs	Particulars	_Rs
To, Balance b/d	50,000	By Bank (sale)	8,500
To P& L A/c	500	By Balance c/d	60,000
To Bank (purchases) (bal/fig)	18,000		
	68,500		68,500